```
vboxuser@ubuntufordevops:~$ ansible --version
ansible [core 2.17.14]
  config file = /etc/ansible/ansible.cfg
  configured module search path = ['/home/vboxuser/.ansible/plugins/modules',
'/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3/dist-packages/ansible
  ansible collection location = /home/vboxuser/.ansible/collections:/usr/share/ansible/collections
  executable location = /usr/bin/ansible
  python version = 3.10.12 (main, Jul 29 2024, 16:56:48) [GCC 11.4.0] (/usr/bin/python3)
  jinja version = 3.0.3
  libyaml = True
vboxuser@ubuntufordevops:~$ ssh-keygen -t ras -C "devops default"
unknown key type ras
vboxuser@ubuntufordevops:~$ ssh-keygen -t rsa -C "devops default"
Generating public/private rsa key pair.
Enter file in which to save the key (/home/vboxuser/.ssh/id_rsa):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/vboxuser/.ssh/id_rsa
Your public key has been saved in /home/vboxuser/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:0gMUTV4QlNLyhTiCSrzPa7ZdVmJ0MJqgO5PD3ofI5jI devops default
The key's randomart image is:
+---[RSA 3072]----+
|.  o  oX==.      |
| oo o.B.B..      |
|.o.  +.*.o       |
|o.o   .oo        |
| Bo   .oS.       |
|o *o. ..o.       |
| = o.. o         |
|E  +o o          |
|.oo...           |
+----[SHA256]-----+
vboxuser@ubuntufordevops:~$ ls -la.ssh
ls: invalid option -- '.'
Try 'ls --help' for more information.
vboxuser@ubuntufordevops:~$ ls -la .ssh
total 16
drwx------  2 vboxuser vboxuser 4096 Mar 23 10:32 .
drwxr-x--- 17 vboxuser vboxuser 4096 Mar 17 11:53 ..
-rw-------  1 vboxuser vboxuser 2602 Mar 23 10:32 id_rsa
-rw-r--r--  1 vboxuser vboxuser  568 Mar 23 10:32 id_rsa.pub
vboxuser@ubuntufordevops:~$ ssh-copy-id -i ~/.ssh/id_rsa.pub"ipaddress"
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```
/usr/bin/ssh-copy-id: ERROR: failed to open ID file
'/home/vboxuser/.ssh/id_rsa.pubipaddress.pub': No such file
vboxuser@ubuntufordevops:~$ ssh-copy-id -i ~/ .ssh/id_rsa.pub "ipaddress"

/usr/bin/ssh-copy-id: ERROR: failed to open ID file '/home/vboxuser/.pub': No such file
        (to install the contents of '/home/vboxuser/.pub' anyway, look at the -f option)
vboxuser@ubuntufordevops:~$ ssh-copy-id -i ~/ .ssh/id_rsa.pub "10.0.2.15"

/usr/bin/ssh-copy-id: ERROR: failed to open ID file '/home/vboxuser/.pub': No such file
        (to install the contents of '/home/vboxuser/.pub' anyway, look at the -f option)
vboxuser@ubuntufordevops:~$ ssh-copy-id -i ~/ .ssh/id_rsa.pub 10.0.2.15

/usr/bin/ssh-copy-id: ERROR: failed to open ID file '/home/vboxuser/.pub': No such file
        (to install the contents of '/home/vboxuser/.pub' anyway, look at the -f option)
vboxuser@ubuntufordevops:~$ ssh-copy-id -i ~/.ssh/id_rsa.pub"10.0.2.15"

/usr/bin/ssh-copy-id: ERROR: failed to open ID file
'/home/vboxuser/.ssh/id_rsa.pub10.0.2.15.pub': No such file
vboxuser@ubuntufordevops:~$ ssh-copy-id -i ~/.ssh/id_rsa.pub "10.0.2.15"
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/vboxuser/.ssh/id_rsa.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are
already installed

/usr/bin/ssh-copy-id: ERROR: ssh: connect to host 10.0.2.15 port 22: Connection refused

vboxuser@ubuntufordevops:~$ sudo apt install openssh-server
[sudo] password for vboxuser:
Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontend. It is heWaiting for cache
lock: Could not get lock /var/lib/dpkg/lock-frontend. It is held by process 2754 (unattended-upgr)
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vboxuser@ubuntufordevops:~$ sudo apt install openssh-server
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vboxuser@ubuntufordevops:~$ sudo apt install openssh-server sudo systemctl restart sshd
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Waiting for cache lock: Could not get lock /var/lib/dpkg/lock-frontend. It ^C held by process 2754
(unattended-upgr)... 40s
vboxuser@ubuntufordevops:~$ ps -p 2754 -o pid,cmd
    PID CMD
   2754 /usr/bin/python3 /usr/bin/unattended-upgrade
vboxuser@ubuntufordevops:~$ sudo kill -9 2754
vboxuser@ubuntufordevops:~$ sudo dpkg --configure -a
sudo apt update
Reading package lists... Done
E: Could not get lock /var/lib/apt/lists/lock. It is held by process 51411 (apt-get)
N: Be aware that removing the lock file is not a solution and may break your system.
E: Unable to lock directory /var/lib/apt/lists/
vboxuser@ubuntufordevops:~$ sudo apt install openssh-server
Reading package lists... Done
```

```
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  libreoffice-ogltrans
Use 'sudo apt autoremove' to remove it.
The following additional packages will be installed:
  ncurses-term openssh-sftp-server ssh-import-id
Suggested packages:
  molly-guard monkeysphere ssh-askpass
The following NEW packages will be installed:
  ncurses-term openssh-server openssh-sftp-server ssh-import-id
0 upgraded, 4 newly installed, 0 to remove and 156 not upgraded.
Need to get 751 kB of archives.
After this operation, 6,050 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 openssh-sftp-server
amd64 1:8.9p1-3ubuntu0.14 [38.9 kB]
Get:2 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 openssh-server amd64
1:8.9p1-3ubuntu0.14 [435 kB]
Get:3 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 ncurses-term all
6.3-2ubuntu0.1 [267 kB]
Get:4 http://in.archive.ubuntu.com/ubuntu jammy/main amd64 ssh-import-id all 5.11-0ubuntu1
[10.1 kB]
Fetched 751 kB in 18s (41.8 kB/s)
Preconfiguring packages ...
Selecting previously unselected package openssh-sftp-server.
(Reading database ... 229850 files and directories currently installed.)
Preparing to unpack .../openssh-sftp-server_1%3a8.9p1-3ubuntu0.14_amd64.deb
 ...
Unpacking openssh-sftp-server (1:8.9p1-3ubuntu0.14) ...
Selecting previously unselected package openssh-server.
Preparing to unpack .../openssh-server_1%3a8.9p1-3ubuntu0.14_amd64.deb ...
Unpacking openssh-server (1:8.9p1-3ubuntu0.14) ...
Selecting previously unselected package ncurses-term.
Preparing to unpack .../ncurses-term_6.3-2ubuntu0.1_all.deb ...
Unpacking ncurses-term (6.3-2ubuntu0.1) ...
Selecting previously unselected package ssh-import-id.
Preparing to unpack .../ssh-import-id_5.11-0ubuntu1_all.deb ...
Unpacking ssh-import-id (5.11-0ubuntu1) ...
Setting up openssh-sftp-server (1:8.9p1-3ubuntu0.14) ...
Setting up openssh-server (1:8.9p1-3ubuntu0.14) ...

Creating config file /etc/ssh/sshd_config with new version
Creating SSH2 RSA key; this may take some time ...
```

```
3072 SHA256:VjOBkZZV5NgUg3DCx4AuRD7olHF93znj2fBvwKzD/7w root@ubuntufordevop
s (RSA)
Creating SSH2 ECDSA key; this may take some time ...
256 SHA256:NKcXcK2QoqR0EQAnrbQLObyud53tGcV0opb6hPGX/b4 root@ubuntufordevops
 (ECDSA)
Creating SSH2 ED25519 key; this may take some time ...
256 SHA256:J7NUK2oLjn21x+WrBP0JO1NcqiJNjDNMLWg3rwJ7Gxk root@ubuntufordevops
 (ED25519)
Created symlink /etc/systemd/system/sshd.service → /lib/systemd/system/ssh.
service.
Created symlink /etc/systemd/system/multi-user.target.wants/ssh.service → /
lib/systemd/system/ssh.service.
rescue-ssh.target is a disabled or a static unit, not starting it.
ssh.socket is a disabled or a static unit, not starting it.
Setting up ssh-import-id (5.11-0ubuntu1) ...
Setting up ncurses-term (6.3-2ubuntu0.1) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for ufw (0.36.1-4ubuntu0.1) ...
vboxuser@ubuntufordevops:~$  sudo systemctl restart sshd
vboxuser@ubuntufordevops:~$ ssh-copy-id -i ~/.ssh/id_rsa.pub "10.0.2.15"
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/vboxuser/.ssh/id_rsa.pub"
The authenticity of host '10.0.2.15 (10.0.2.15)' can't be established.
ED25519 key fingerprint is SHA256:J7NUK2oLjn21x+WrBP0JO1NcqiJNjDNMLWg3rwJ7Gxk.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are
already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to
install the new keys
vboxuser@10.0.2.15's password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh '10.0.2.15'"
and check to make sure that only the key(s) you wanted were added.

vboxuser@ubuntufordevops:~$ - ssh "10.0.2.15"
-: command not found
vboxuser@ubuntufordevops:~$ -ssh "10.0.2.15"
Command '-ssh' not found, did you mean:
  command 'mssh' from deb mssh (2.2-5)
  command 'cssh' from deb clusterssh (4.16-3)
  command 'sssh' from deb guile-ssh (0.13.1-6)
  command 'bssh' from deb avahi-ui-utils (0.8-5ubuntu5.4)
```

```
  command 'ssh' from deb openssh-client (1:8.9p1-3ubuntu0.14)
  command 'zssh' from deb zssh (1.5c.debian.1-8)
Try: sudo apt install <deb name>
vboxuser@ubuntufordevops:~$ - ssh"10.0.2.15"
-: command not found
vboxuser@ubuntufordevops:~$  ssh"10.0.2.15"
ssh10.0.2.15: command not found
vboxuser@ubuntufordevops:~$ ssh "10.0.2.15"
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-106-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

156 updates can be applied immediately.
61 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

2 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

*** System restart required ***
vboxuser@ubuntufordevops:~$ sudo nano inventory
[sudo] password for vboxuser:
vboxuser@ubuntufordevops:~$ ssh "10.0.2.15"
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-106-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

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*** System restart required ***
Last login: Mon Mar 23 10:59:57 2026 from 10.0.2.15
```

```
vboxuser@ubuntufordevops:~$ inventory ansible all --ket-file ~/.ssh/id_rsa -i
inventory: command not found
vboxuser@ubuntufordevops:~$ sudo nano playbook.yaml
[sudo] password for vboxuser:
vboxuser@ubuntufordevops:~$ ansible-playbook -i inventory playbook.yaml

PLAY [Echo] ********************************************************************

TASK [Gathering Facts] *********************************************************
[WARNING]: Platform linux on host 10.0.2.15 is using the discovered Python
interpreter at /usr/bin/python3.10, but future installation of another Python
interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
ok: [10.0.2.15]

TASK [Print debug message] *****************************************************
ok: [10.0.2.15] => {
    "msg": "Hello, world!"
}

PLAY RECAP *********************************************************************
10.0.2.15                  : ok=2  changed=0    unreachable=0    failed=0    skipped=0
rescued=0    ignored=0

vboxuser@ubuntufordevops:~$ ansible all --key-file~/.ssh/id_rsa -i inventory -m ping
usage: ansible [-h] [--version] [-v] [-b] [--become-method BECOME_METHOD]
    [--become-user BECOME_USER]
    [-K | --become-password-file BECOME_PASSWORD_FILE]
    [-i INVENTORY] [--list-hosts] [-l SUBSET]
    [-P POLL_INTERVAL] [-B SECONDS] [-o] [-t TREE]
    [--private-key PRIVATE_KEY_FILE] [-u REMOTE_USER]
    [-c CONNECTION] [-T TIMEOUT]
    [--ssh-common-args SSH_COMMON_ARGS]
    [--sftp-extra-args SFTP_EXTRA_ARGS]
    [--scp-extra-args SCP_EXTRA_ARGS]
    [--ssh-extra-args SSH_EXTRA_ARGS]
    [-k | --connection-password-file CONNECTION_PASSWORD_FILE]
    [-C] [-D] [-e EXTRA_VARS] [--vault-id VAULT_IDS]
    [-J | --vault-password-file VAULT_PASSWORD_FILES]
    [-f FORKS] [-M MODULE_PATH] [--playbook-dir BASEDIR]
    [--task-timeout TASK_TIMEOUT] [-a MODULE_ARGS]
    [-m MODULE_NAME]
    pattern
```

```
ansible: error: unrecognized arguments: --key-file~/.ssh/id_rsa

usage: ansible [-h] [--version] [-v] [-b] [--become-method BECOME_METHOD]
           [--become-user BECOME_USER]
           [-K | --become-password-file BECOME_PASSWORD_FILE]
           [-i INVENTORY] [--list-hosts] [-l SUBSET]
           [-P POLL_INTERVAL] [-B SECONDS] [-o] [-t TREE]
           [--private-key PRIVATE_KEY_FILE] [-u REMOTE_USER]
           [-c CONNECTION] [-T TIMEOUT]
           [--ssh-common-args SSH_COMMON_ARGS]
           [--sftp-extra-args SFTP_EXTRA_ARGS]
           [--scp-extra-args SCP_EXTRA_ARGS]
           [--ssh-extra-args SSH_EXTRA_ARGS]
           [-k | --connection-password-file CONNECTION_PASSWORD_FILE]
           [-C] [-D] [-e EXTRA_VARS] [--vault-id VAULT_IDS]
           [-J | --vault-password-file VAULT_PASSWORD_FILES]
           [-f FORKS] [-M MODULE_PATH] [--playbook-dir BASEDIR]
           [--task-timeout TASK_TIMEOUT] [-a MODULE_ARGS]
           [-m MODULE_NAME]
           pattern

Define and run a single task 'playbook' against a set of hosts

positional arguments:
  pattern                 host pattern

options:
  --become-password-file BECOME_PASSWORD_FILE, --become-pass-file
BECOME_PASSWORD_FILE
                    Become password file
  --connection-password-file CONNECTION_PASSWORD_FILE, --conn-pass-file
CONNECTION_PASSWORD_FILE
                    Connection password file
  --list-hosts            outputs a list of matching hosts; does not
                    execute anything else
  --playbook-dir BASEDIR
                    Since this tool does not use playbooks, use this
                    as a substitute playbook directory. This sets the
                    relative path for many features including roles/
                    group_vars/ etc.
  --task-timeout TASK_TIMEOUT
                    set task timeout limit in seconds, must be
                    positive integer.
  --vault-id VAULT_IDS  the vault identity to use. This argument may be
```

```
                    specified multiple times.
  --vault-password-file VAULT_PASSWORD_FILES, --vault-pass-file
VAULT_PASSWORD_FILES
                    vault password file
  --version                 show program's version number, config file
                    location, configured module search path, module
                    location, executable location and exit
  -B SECONDS, --background SECONDS
                    run asynchronously, failing after X seconds
                    (default=N/A)
  -C, --check               don't make any changes; instead, try to predict
                    some of the changes that may occur
  -D, --diff        when changing (small) files and templates, show
                    the differences in those files; works great with
                    --check
  -J, --ask-vault-password, --ask-vault-pass
                    ask for vault password
  -K, --ask-become-pass
                    ask for privilege escalation password
  -M MODULE_PATH, --module-path MODULE_PATH
                    prepend colon-separated path(s) to module library
                    (default={{ ANSIBLE_HOME ~ "/plugins/modules:/usr
                    /share/ansible/plugins/modules" }}). This
                    argument may be specified multiple times.
  -P POLL_INTERVAL, --poll POLL_INTERVAL
                    set the poll interval if using -B (default=15)
  -a MODULE_ARGS, --args MODULE_ARGS
                    The action's options in space separated k=v
                    format: -a 'opt1=val1 opt2=val2' or a json
                    string: -a '{"opt1": "val1", "opt2": "val2"}'
  -e EXTRA_VARS, --extra-vars EXTRA_VARS
                    set additional variables as key=value or
                    YAML/JSON, if filename prepend with @. This
                    argument may be specified multiple times.
  -f FORKS, --forks FORKS
                    specify number of parallel processes to use
                    (default=5)
  -h, --help                show this help message and exit
  -i INVENTORY, --inventory INVENTORY, --inventory-file INVENTORY
                    specify inventory host path or comma separated
                    host list. --inventory-file is deprecated. This
                    argument may be specified multiple times.
  -k, --ask-pass            ask for connection password
  -l SUBSET, --limit SUBSET
```

```
                    further limit selected hosts to an additional
                    pattern
  -m MODULE_NAME, --module-name MODULE_NAME
                    Name of the action to execute (default=command)
  -o, --one-line            condense output
  -t TREE, --tree TREE  log output to this directory
  -v, --verbose             Causes Ansible to print more debug messages.
                    Adding multiple -v will increase the verbosity,
                    the builtin plugins currently evaluate up to
                    -vvvvvv. A reasonable level to start is -vvv,
                    connection debugging might require -vvvv. This
                    argument may be specified multiple times.

Privilege Escalation Options:
  control how and which user you become as on target hosts

  --become-method BECOME_METHOD
                    privilege escalation method to use
                    (default=sudo), use `ansible-doc -t become -l` to
                    list valid choices.
  --become-user BECOME_USER
                    run operations as this user (default=root)
  -b, --become              run operations with become (does not imply
                    password prompting)

Connection Options:
  control as whom and how to connect to hosts

  --private-key PRIVATE_KEY_FILE, --key-file PRIVATE_KEY_FILE
                    use this file to authenticate the connection
  --scp-extra-args SCP_EXTRA_ARGS
                    specify extra arguments to pass to scp only (e.g.
                    -l)
  --sftp-extra-args SFTP_EXTRA_ARGS
                    specify extra arguments to pass to sftp only
                    (e.g. -f, -l)
  --ssh-common-args SSH_COMMON_ARGS
                    specify common arguments to pass to sftp/scp/ssh
                    (e.g. ProxyCommand)
  --ssh-extra-args SSH_EXTRA_ARGS
                    specify extra arguments to pass to ssh only (e.g.
                    -R)
  -T TIMEOUT, --timeout TIMEOUT
                    override the connection timeout in seconds
```

(default depends on connection)
  -c CONNECTION, --connection CONNECTION
                connection type to use (default=ssh)
  -u REMOTE_USER, --user REMOTE_USER
                connect as this user (default=None)

Some actions do not make sense in Ad-Hoc (include, meta, etc)
vboxuser@ubuntufordevops:~$ ansibleall --key-file~/.ssh/id_rsa -i inventory -m ping
ansibleall: command not found
vboxuser@ubuntufordevops:~$ ansible all --key-file ~/.ssh/id_rsa -i inventory -m ping
[WARNING]: Platform linux on host 10.0.2.15 is using the discovered Python interpreter at /usr/bin/python3.10, but future installation of another Python interpreter could change the meaning of that path. See https://docs.ansible.com/ansible-core/2.17/reference_appendices/interpreter_discovery.html for more information.
10.0.2.15 | SUCCESS => {
    "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3.10"
    },
    "changed": false,
    "ping": "pong"
}
vboxuser@ubuntufordevops:~$

```
vboxuser@ubuntufordevops:~$ ansible --version
ansible [core 2.17.14]
  config file = /etc/ansible/ansible.cfg
  configured module search path = ['/home/vboxuser/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3/dist-packages/ansible
  ansible collection location = /home/vboxuser/.ansible/collections:/usr/share/ansible/collections
  executable location = /usr/bin/ansible
  python version = 3.10.12 (main, Jul 29 2024, 16:56:48) [GCC 11.4.0] (/usr/bin/python3)
  jinja version = 3.0.3
  libyaml = True
```

aaaa

```
vboxuser@ubuntufordevops:~$ ssh-keygen -t rsa -C "devops default"
Generating public/private rsa key pair.
Enter file in which to save the key (/home/vboxuser/.ssh/id_rsa):
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/vboxuser/.ssh/id_rsa
Your public key has been saved in /home/vboxuser/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:0gMUTV4QlNLyhTiCSrzPa7ZdVmJ0MJqgO5PD3ofI5jI devops default
The key's randomart image is:
+---[RSA 3072]----+
|.  o  oX==.      |
| oo o.B.B..      |
|.o.  +.*.o       |
|o.o   .oo        |
| Bo   .oS.       |
|o *o. ..o.       |
| = o.. o         |
|E  +o o          |
|.oo...           |
+----[SHA256]-----+
```

aaaaa

```
vboxuser@ubuntufordevops:~$ ls -la .ssh
total 16
drwx------  2 vboxuser vboxuser 4096 Mar 23 10:32 .
drwxr-x--- 17 vboxuser vboxuser 4096 Mar 17 11:53 ..
-rw-------  1 vboxuser vboxuser 2602 Mar 23 10:32 id_rsa
-rw-r--r--  1 vboxuser vboxuser  568 Mar 23 10:32 id_rsa.pub
```

aaa

```
vboxuser@ubuntufordevops:~$ sudo apt install openssh-server
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  libreoffice-ogltrans
Use 'sudo apt autoremove' to remove it.
The following additional packages will be installed:
  ncurses-term openssh-sftp-server ssh-import-id
Suggested packages:
  molly-guard monkeysphere ssh-askpass
The following NEW packages will be installed:
  ncurses-term openssh-server openssh-sftp-server ssh-import-id
0 upgraded, 4 newly installed, 0 to remove and 156 not upgraded.
Need to get 751 kB of archives.
After this operation, 6,050 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 openssh-sftp-server amd64 1:8.9p1-3ubuntu0.14 [38.9 kB]
Get:2 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 openssh-server amd64 1:8.9p1-3ubuntu0.14 [435 kB]
Get:3 http://in.archive.ubuntu.com/ubuntu jammy-updates/main amd64 ncurses-term all 6.3-2ubuntu0.1 [267 kB]
Get:4 http://in.archive.ubuntu.com/ubuntu jammy/main amd64 ssh-import-id all 5.11-0ubuntu1 [10.1 kB]
Fetched 751 kB in 18s (41.8 kB/s)
Preconfiguring packages ...
Selecting previously unselected package openssh-sftp-server.
(Reading database ... 229850 files and directories currently installed.)
Preparing to unpack .../openssh-sftp-server_1%3a8.9p1-3ubuntu0.14_amd64.deb
 ...
Unpacking openssh-sftp-server (1:8.9p1-3ubuntu0.14) ...
Selecting previously unselected package openssh-server.
Preparing to unpack .../openssh-server_1%3a8.9p1-3ubuntu0.14_amd64.deb ...
Unpacking openssh-server (1:8.9p1-3ubuntu0.14) ...
Selecting previously unselected package ncurses-term.
Preparing to unpack .../ncurses-term_6.3-2ubuntu0.1_all.deb ...
Unpacking ncurses-term (6.3-2ubuntu0.1) ...
Selecting previously unselected package ssh-import-id.
Preparing to unpack .../ssh-import-id_5.11-0ubuntu1_all.deb ...
Unpacking ssh-import-id (5.11-0ubuntu1) ...
Setting up openssh-sftp-server (1:8.9p1-3ubuntu0.14) ...
Setting up openssh-server (1:8.9p1-3ubuntu0.14) ...

Creating config file /etc/ssh/sshd_config with new version
Creating SSH2 RSA key; this may take some time ...
3072 SHA256:VjOBkZZV5NgUg3DCx4AuRD7olHF93znj2fBvwKzD/7w root@ubuntufordevop
s (RSA)
Creating SSH2 ECDSA key; this may take some time ...
256 SHA256:NKcXcKZQoqR0EQAnrbQLObyud53tGcV0opb6hPGX/b4 root@ubuntufordevops
 (ECDSA)
Creating SSH2 ED25519 key; this may take some time ...
256 SHA256:J7NUK2oLjn21x+WrBP0JO1NcqiJNjDNMLWg3rwJ7Gxk root@ubuntufordevops
 (ED25519)
Created symlink /etc/systemd/system/sshd.service → /lib/systemd/system/ssh.
service.
Created symlink /etc/systemd/system/multi-user.target.wants/ssh.service → /
lib/systemd/system/ssh.service.
rescue-ssh.target is a disabled or a static unit, not starting it.
ssh.socket is a disabled or a static unit, not starting it.
Setting up ssh-import-id (5.11-0ubuntu1) ...
Setting up ncurses-term (6.3-2ubuntu0.1) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for ufw (0.36.1-4ubuntu0.1) ...
```

aaaa

```
vboxuser@ubuntufordevops:~$  sudo systemctl restart sshd
vboxuser@ubuntufordevops:~$ ssh-copy-id -i ~/.ssh/id_rsa.pub "10.0.2.15"
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/vboxuser/.ssh/id_rsa.pub"
The authenticity of host '10.0.2.15 (10.0.2.15)' can't be established.
ED25519 key fingerprint is SHA256:J7NUK2oLjn21x+WrBP0JO1NcqiJNjDNMLWg3rwJ7Gxk.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to install the new keys
vboxuser@10.0.2.15's password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh '10.0.2.15'"
and check to make sure that only the key(s) you wanted were added.
```

aaaaa

```
vboxuser@ubuntufordevops:~$ sudo nano inventory
[sudo] password for vboxuser:
vboxuser@ubuntufordevops:~$ ssh "10.0.2.15"
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-106-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

156 updates can be applied immediately.
61 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

2 additional security updates can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

*** System restart required ***
Last login: Mon Mar 23 10:59:57 2026 from 10.0.2.15
```

aaaaaa

```
  GNU nano 6.2
[servers]
10.0.2.15
```

Aaaa

```
vboxuser@ubuntufordevops:~$ sudo nano playbook.yaml
[sudo] password for vboxuser:
vboxuser@ubuntufordevops:~$ ansible-playbook -i inventory playbook.yaml

PLAY [Echo] *******************************************************************

TASK [Gathering Facts] ********************************************************
[WARNING]: Platform linux on host 10.0.2.15 is using the discovered Python
interpreter at /usr/bin/python3.10, but future installation of another Python
interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
ok: [10.0.2.15]

TASK [Print debug message] ****************************************************
ok: [10.0.2.15] => {
    "msg": "Hello, world!"
}

PLAY RECAP *********************************************************************
10.0.2.15                  : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

aaaa

```
  GNU nano 6.2                                                    playbook.yaml
---
- name: Echo
  hosts: servers
  connection: local
  tasks:
    - name: Print debug message
      debug:
        msg: Hello, world!
```

aa

```
vboxuser@ubuntufordevops:~$ ansible all --key-file ~/.ssh/id_rsa -i inventory -m ping
[WARNING]: Platform linux on host 10.0.2.15 is using the discovered Python
interpreter at /usr/bin/python3.10, but future installation of another Python
interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
10.0.2.15 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3.10"
    },
    "changed": false,
    "ping": "pong"
}
```

aaaaa

```
vboxuser@ubuntufordevops:~$ ansible-playbook -i inventory playbook.yaml

PLAY [Echo] *******************************************************************

TASK [Gathering Facts] *********************************************************
[WARNING]: Platform linux on host 10.0.2.15 is using the discovered Python
interpreter at /usr/bin/python3.10, but future installation of another Python
interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-
core/2.17/reference_appendices/interpreter_discovery.html for more information.
ok: [10.0.2.15]

TASK [Print debug message] *****************************************************
ok: [10.0.2.15] => {
    "msg": "Hello, world!"
}

PLAY RECAP *********************************************************************
10.0.2.15                  : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

aaaa